

Molecular mapping and candidate gene analysis of a new epicuticular wax locus in sorghum (*Sorghum bicolor* L. Moench)

G. Anurag Uttam¹ · M. Praveen¹ · Y. Venkateswara Rao² · Vilas A. Tonapi¹ · R. Madhusudhana¹ 

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Abstract

Key message A new epicuticular wax (bloom) locus has been identified and fine mapped to the 207.89 kb genomic region on chromosome 1. A putative candidate gene, Sobic.001G269200, annotated as GDSL-like lipase/acylhydrolase, is proposed as the most probable candidate gene involved in bloom synthesis/deposition.

Abstract Deposition of epicuticular wax on plant aerial surface is one strategy that plants adapt to reduce non-transpiration water loss. Epicuticular wax (bloom)-less mutants in sorghum with their glossy phenotypes exhibit changes in the accumulation of epicuticular wax on leaf and culm surfaces. We report molecular mapping of a new sorghum locus, bloomless mutant (bm39), involved in epicuticular wax biosynthesis in sorghum. Inheritance studies involving a profusely bloom parent (BTx623) and a spontaneous bloomless mutant (RS647) indicated that the parents differed in a single gene for bloom synthesis. Bloomless was recessive to bloom deposition. Genetic mapping involving F₂ and F₇ mapping populations in diverse genetic backgrounds (BTx623 × RS647; 296A × RS647 and

27A × RS647) identified and validated the map location of bm39 to a region of 207.89 kb on chromosome 1. SSR markers, Sblm13 and Sblm16, flanked the bm39 locus to a map interval of 0.3 cM on either side. Nine candidate genes were identified, of which Sobic.001G269200 annotated for GDSL-like lipase/acylhydrolase is the most likely gene associated with epicuticular wax deposition. Gene expression analysis in parents, isogenic lines and sets of near isogenic lines also confirmed the reduced expression of the putative candidate gene. The study opens possibilities for a detailed molecular analysis of the gene, its role in epicuticular wax synthesis and deposition, and may help to understand its function in moisture stress tolerance and insect and pathogen resistance in sorghum.

Abbreviations

CMS	Cytoplasmic male sterility
CTAB	Cetyltrimethylammonium bromide
dNTP	Deoxynucleotide triphosphate
GBSS	Granule-bound starch synthase
GDSL	Glycine aspartic (glutamic) serine leucine
ORF	Open reading frame
PCR	Polymerase chain reaction
RT-PCR	Reverse transcriptase polymerase chain reaction
SNP	Single nucleotide polymorphism
SSRs	Simple sequence repeats

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✉ R. Madhusudhana
madhu@millets.res.in

¹ Marker-Assisted Selection Laboratory, ICAR-Indian Institute of Millet Research (IIMR), Rajendranagar, Hyderabad 500 030, India

² Department of Botany, Andhra University, Visakhapatnam, Andhra Pradesh, India

Introduction

Sorghum (*Sorghum bicolor* L. Moench) is a highly productive C4 cereal crop that is well adapted to arid and semi-arid environments (Rooney et al. 2007). It ranks fifth in acreage among the cereals after maize, rice, wheat and barley (FAO 2015). Besides being a major source of staple

food for millions of people in arid and semi-arid regions, it serves as an important source of feed and fodder for animals. Sorghum requires less water and can grow well in moisture stress conditions and in hotter regions that makes the sorghum plant superior to maize, which has almost similar seasonal requirements. In recent times, sorghum has emerged as a potential alternative bio-energy crop due to increasing demand for fuel-grade ethanol (Calvino 2012; Sambusiti et al. 2013).

Plants are exposed to various biotic and abiotic stresses that directly or indirectly affect their growth and development (Pasha et al. 2015). Abiotic stress arises from exposure to climatic extremes such as drought, heat and cold. To survive moisture stress, the plant must have several traits that help it to either extract more water from the soil or effectively control the amount of water loss through transpiration and other losses. Some of the traits that are known to play a role in drought tolerance include leaf conductance, osmotic adjustment, leaf rolling, leaf water potential, proline accumulation, epicuticular wax and root growth (Rosenow et al. 1983). Control over non-stomatal water loss from leaves is one of the key components contributing to drought tolerance. Epicuticular wax (EW) is one such special component developed by plants to seal and protect the aerial organs from various biotic and abiotic stresses (Barozai and Husnain 2014).

In sorghum, EW deposition as white soft material (bloom) starts appearing at the seedling stage. However, its deposition is very prominent at the boot leaf stage (Jenks et al. 2000; Jordan et al. 1983) on the leaf sheath and the lower part of the leaf blade (Bianchi et al. 1978). Sorghum produces a higher amount of bloom than rice and corn on per area and weight basis (Beattie and Marcell 2002; Burow et al. 2008; O'Toole and Cruz 1983). Bloom is associated with several traits including grain yield in sorghum. Studies have shown that bloom lines yield significantly higher than their bloomless isogenic lines (Ross 1972; Webster and Schmalzel 1979), which was attributed to higher gas exchange ratio in bloom lines (Chatterton et al. 1975). It preserves water status by reducing the solar energy load on the leaf surface and improves water use efficiency by regulating nighttime water loss (Burow et al. 2008; Premachandra et al. 1994), cuticular transpiration and enhanced stomatal control of water loss (Blum 1975; Jordan et al. 1984). In vitro and in vivo dry matter digestibility of forage in bloomless lines was superior by 22% over bloom lines (Cummins and Dobson 1972; Cummins and Sudweeks 1976). Epicuticular wax also plays important roles in plant defense against bacterial and fungal pathogens and impacts on plant–insect interactions. Bloom in sorghum is associated with resistance to green bug (Weibel and Starks 1986) and fungal pathogen, *Exserohilum turcicum* (Jenks et al. 1994), sheath blight (Kasuga et al. 2001)

and forage digestibility (Cummins and Sudweeks 1976). Linkage between a major plant height gene and bloom locus is also reported in sorghum (Madhusudhana and Patil 2013). These studies indicate that the EW is key to cuticular transpiration and could be involved in pleiotropic expression in many other traits.

Plant cuticular wax biosynthesis and its loading to the plant surface was reported in plants such as rice, wheat, maize, barley, *Brassica* and *Arabidopsis*. (Holloway et al. 1977; Jenks et al. 2002; Meeks et al. 2012; O'Toole and Cruz 1983; Uddin and Marshall 1988; Zhang 2012). Mutants with reduced wax deposition or altered wax composition are commonly characterized by a bright green phenotype that can be detected visually (Aarts et al. 1995). Wax-deficient mutants have been reported in other plants, including maize, sorghum, barley and rape (Kunst and Samuels 2003). Studies of the eceriferum (*cer*) mutants in *Arabidopsis* and glossy (*gl*) mutants in maize and rice led to the identification and isolation of a number of wax-related genes (Gan et al. 2016; Li et al. 2013; Samuels et al. 2008). In wheat, the wax on leaves and stems was mainly controlled by two sets of genes: glaucousness loci (*W1* and *W2*) and non-glaucousness loci (*Iw1* and *Iw2*) (Lu et al. 2015). Sorghum is one of the crops where molecular information on genes involved in epicuticular wax synthesis needs to be largely uncovered. Several bloomless mutants and their allelic relations have been reported in recent studies (Burow et al. 2008; Jenks et al. 1994; Peters et al. 2009; Punnuri et al. 2017). Of the several bloomless mutants reported, only the map position of Bloom Cuticle (*BLMC*) locus was reported on SBI-10 and was predicted to contain an acyl-CoA oxidase (Sb10g000390 = Sobic.010G001900) and seven other putative transcripts (Burow et al. 2009). *BLMC* was found to be allelic to *bm2*, could have pleiotropic effects, including cuticle proper synthesis, leaf water retention, greenbug biotype C (*Schizaphis graminum* Rondani) susceptibility and resistance to the fungal pathogen *E. turcicum* Pass., which causes northern corn leaf blight (Punnuri et al. 2017). Two other recent studies have also identified Sb06g023280 (Sobic.006G155700) (=AT1G17840 (WBC11) that encodes an ABC transporter responsible for wax secretion in *Arabidopsis*) (Mizuno et al. 2013), and Sobic.009G083300 (=AT1G51500 (CER5), like white-brown complex homolog protein 12) and Sobic.001G453200 (=AT1G68530 (CER6), similar to very-long-chain fatty acid condensing enzyme, putative, expressed) genes to be involved in bloom production and deposition in sorghum (Jiao et al. 2016).

Elucidation of the molecular basis of the epicuticular wax pathway and cuticular function as done in model crops like *Arabidopsis* is essential in sorghum, a model cereal crop known for its profuse epicuticular wax deposition and drought tolerance. Toward complementing the knowledge

on epicuticular wax genes in sorghum, we undertook a study to characterize a bloomless mutant for its inheritance and to identify and validate its molecular map position. Though epicuticular wax is a quantitative trait with many genes involved in its biosynthesis, we undertook a strategy to map a major locus that caused the absence of epicuticular wax (bloomless) in a natural bloomless mutant line, RS647, available in our sorghum program. Linkage analysis indicated the presence of a locus on chromosome 1 involved in the expression of bloom. In line with earlier bloom mutants, we designated the present locus as *bm39*. Our fine-mapping studies identified a putative candidate gene for the synthesis of epicuticular wax.

Materials and methods

Plant material

Two sorghum lines, BTx623 and RS647, were used as parents to develop an F_2 mapping population to characterize and map epicuticular wax (bloom) locus on the sorghum genome. The population was advanced to later generations following the single seed descent (SSD) method until F_7 generation, and a recombinant inbred line (RIL) population consisting of 116 lines was established. Parent BTx623, an elite, white-seeded, inbred line with the pedigree BTx3197/SC170-6-4 developed by Dr. Fred Miller at Texas A&M University (Burow et al. 2011), produces significant amount of bloom on its leaf and culm surface (Fig. 1). This inbred line was used as a parent in several genetic studies and for genetic map construction (Bowers et al. 2003; Chitenden et al. 1994; Kong and Hart 2000; Menz et al. 2002).

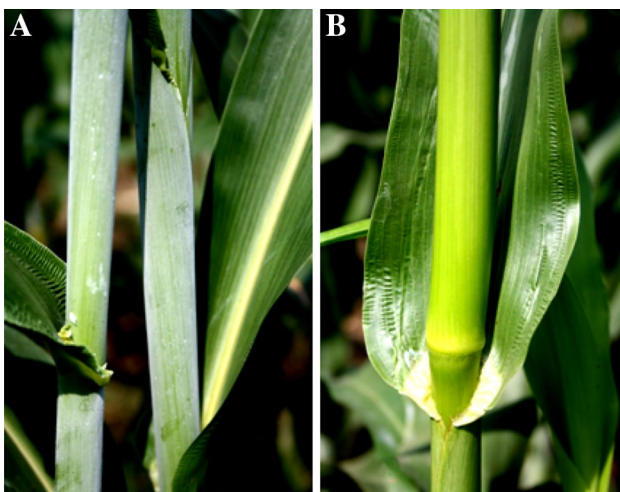


Fig. 1 Photographs of **a** BTx623 with profuse bloom and **b** RS647 with absence of bloom used in the study. The photographs show the culm part of the leaf and sheath of the internode at anthesis

The whole genome of BTx623 was sequenced (Paterson et al. 2009) and is publicly available (https://phytozome.jgi.doe.gov/pz/portal.html#!info?alias=Org_Sbicolor). The other parent, RS647, is a spontaneous bloomless mutant (with no visible epicuticular wax load, and no other agronomic abnormalities), developed from a cross between IS15703 $\times$ RS29 during an earlier breeding program at the ICAR-Indian Institute of Millets Research (ICAR-IIMR) Hyderabad, India. Apart from BTx623, two Indian CMS lines with profuse bloom, 27A and 296A, were also crossed with bloomless RS647 to develop two F_2 segregating populations for validation of the results from the BTx623 $\times$ RS647 cross.

Using BTx623 as the donor for the bloom trait, the bloomless parent RS647 was converted to its isogenic bloom version through repeated backcrossing for over five generations. This converted genotype is hereafter represented as RS647C and was used in epicuticular wax estimation and RNA expression analysis. Lines segregating for bloom and bloomless phenotype in advanced generation like F_6 were identified as heterogeneous inbred families (HIFs) in BTx623 $\times$ RS647 crosses. Plants differing in bloom trait in each HIF were harvested separately and evaluated in subsequent generations as NILs. HIFs, developed from selfing and selection scheme (Allard 1999) that segregate only for a small portion of the genome of interest have been utilized as the source of NILs for rust resistance in common bean (Haley et al. 1994) and peanut (Yeri et al. 2014) and seed weight in sorghum (Tuinstra et al. 1997). Five pairs of NILs were used in epicuticular wax estimation and two pairs of NILs were used in RNA expression analysis along with the two parents BTx623, RS647 and the isogenic converted RS647C.

Phenotypic evaluation

Three F_1 s (P1:BTx623 $\times$ RS647, P2:27A $\times$ RS647 and P3:296A $\times$ RS647) were developed during the post-rainy season of 2011 at ICAR-IIMR. The F_1 plants were grown in cement pots during the rainy season of 2012 and were selfed using paper bags to aid in self-fertilization and avoid cross-pollination. Selfed seed from three F_1 plants (P1 $F_{2:0}$, P2 $F_{2:0}$ and P3 $F_{2:0}$) were harvested separately for further experimentation. F_2 populations were separately raised in the experimental fields of ICAR-IIMR, Hyderabad, during the post-rainy season of 2012–2013. Plants were grown as a single plant per hill with a spacing of 15 cm between plants and 60 cm between rows. Recommended agronomic practices were followed to raise a good crop. One hundred and sixty-three F_2 plants were available in P1 F_2 , whereas P2 F_2 and P3 F_2 had 202 and 240 F_2 plants respectively. Before anthesis, each F_2 plant of the three F_2 populations was labeled individually. At boot leaf stage, F_2 plants were

phenotypically scored on a binary scale for the presence/absence of bloom on the leaf and culm surface as reported (Burow et al. 2009; Punnuri et al. 2017). A plant with bloom load was scored as “B” (bloom) phenotype, while those showing the absence of bloom load were categorized as “BL” (bloomless). The goodness of fit between the observed and expected segregation for bloom and bloomless plants in the F_2 generation of each cross was tested using Chi-square (χ^2) test to understand the genetic control of the bloom trait.

One hundred and sixteen F_7 RIL families of cross BTx623 $\times$ RS647 were available and were evaluated for bloom trait during the post-rainy season of 2015–2016. Each F_7 family was grown as one row plot of 4 m length and around 25 plants within a family were observed for the expression of bloom trait at the boot leaf stage in a similar way as detailed above for the F_2 populations. The goodness of fit between the observed and expected for the bloom and bloomless families in the F_7 generation was tested using Chi-square (χ^2) test.

Quantification of epicuticular wax

At the boot leaf stage, bloom load was estimated on the second top leaf of parents BTx623 (B), RS647 (BL), RS647C (B) and five sets of NILs [11-1(B) and 11-2 (BL); 38-1(B) and 38-2 (BL); 94-1(B) and 94-2 (BL); 102-1(B) and 102-2 (BL); 116-1(B) and 116-2 (BL)]. For wax estimation, leaf samples were collected at weekly intervals for all the genotypes over 6 weeks. Epicuticular wax was extracted from both abaxial and adaxial surfaces using the colorimetric method (Ebercon et al. 1977). Briefly, fresh leaves were cut into pieces of leaf tissue of total area of 30 cm² (both surfaces). Each sample was immersed in 15 ml redistilled chloroform for 15 s. The leaf disc was removed by filtering the solvent and the collected solvent was boiled at 95 °C in a boiling water bath until the chloroform smell could not be detected. After adding 5 ml of reagent (prepared with 20 g powdered potassium dichromate dissolved in 40 ml deionized water and makeup volume to 1 l with concentrated sulfuric acid), the samples were placed in a water bath for 30 min. After cooling, 12 ml of deionized water was added to each sample. Color development was allowed for the samples and then the optical densities were recorded at 590 nm. The readings were converted into wax quantities (mg cm⁻²) using the standard curve prepared with carnauba wax as substrate.

Pattern of water loss

A small experiment was conducted to study the pattern of water loss from the leaf surface of bloomless RS647 and its cognate bloom RS647C. Leaf samples of 30 cm² area from

the second top leaf were collected in three replicates from both lines (RS647 and RS647C) and were immediately measured for their fresh weight (g). Later, the samples were air-dried at room temperature and the weight of the samples were recorded at 15 min interval for a period of 1 h. The mean values over replicates were calculated and water loss per unit time was estimated.

DNA extraction, PCR and electrophoresis

Leaf samples from fresh young leaf tissues of 2–3 weeks' age were collected separately and frozen for genomic DNA extraction. DNA was extracted following the CTAB method (Saghai-Marooft et al. 1984). The quality of the DNA was analyzed by 1% agarose gel electrophoresis. Veriti PCR system (M/s Applied Bio system, USA) was used for PCR amplification with the following buffer composition: about 2 μ l (i.e. 50–100 ng) of template DNA was added to the wells in a PCR plate. The master mix consisted of 2 mM forward primer, 2 mM reverse primer, 1.25 mM dNTPs (M/s Bangalore Genei Pvt. Ltd.), 1 U Taq DNA polymerase (M/s Bangalore Genei Pvt. Ltd.) and 1 $\times$ PCR buffer (Tris with 1.5 mM MgCl₂), and the total volume was made up to 10 μ l using sterile distilled water. The master mix (8.0 μ l) was dispensed to the PCR plate containing 2 μ l template DNA. The thermal profile followed was: initial denaturation at 94 °C for 5 min, followed by 35 cycles of denaturation for 1 min at 94 °C, primer annealing for 30 s, primer extension for 30 s at 72 °C and final extension at 72 °C for 10 min. The amplified products were loaded in 4% agarose gel for high resolving products and in 6% PAGE for low resolving products.

Marker analysis

At first, we surveyed polymorphism using the ten SSRs (Xsbarslbk series) around the bloom locus already mapped (*BLMC*) in sorghum (Burow et al. 2009). The details of the reported *BLMC* linked markers are given in Supplementary Table 1. Secondly, we also selected 240 SSRs of Xtxp, Xgap, Xcup, and SB series (Bhatramakki et al. 2000; Brown et al. 1996; Burow et al. 2009; Schloss et al. 2002; Yonemaru et al. 2009) covering most of the genome for polymorphism survey between the parental genotypes (BTx623 and RS647) (Supplementary Table 2).

Targeted mapping of bloom locus

To identify linkage between marker and bloom locus, we followed bulked segregant analysis (BSA) (Michelmore et al. 1991). For BSA, two bulks, one each from bloom and bloomless plants, were formed in F_2 mapping population as described (Michelmore et al. 1991). For each bulk,

equal quantity of DNA (25 ng μl^{-1}) was taken from 12 individual plants and pooled. The polymorphic SSR markers identified between the parents (BTx623 and RS647) were used in BSA to genotype with parents along with bloom and bloomless bulks. Based on banding pattern in bloom and bloomless bulks and in two parents, the association of SSR markers with locus controlling the target trait was presumed. As indicated by Michelmore et al. (1991), the probability of an unlinked locus being polymorphic between bulks of ten individuals is 2×10^{-6} . Thus, the chances of detecting an unlinked locus as significant in BSA is extremely small.

Additional experiments were conducted to validate the marker linkage of bloom locus identified using BSA. F_2 plants of P_1F_2 , P_2F_2 , P_3F_2 and F_7 RILs were genotyped with significant markers of BSA that were likely to be linked with the bloom locus. Since the two parents of the present study differ for bloom phenotype (RS647 is bloomless, BTx623 is bloom), the bloom morphology was also scored in the F_2 and RIL populations on binary scale and inserted as a morphological marker (bm39) on the linkage map. Local linkage maps around the bloom locus were constructed for three F_2 mapping populations. Seeds from each F_2 plant was planted to establish F_3 progeny rows to identify F_2 plants homozygous for the dominant bloom (bm39). F_3 progeny rows (minimum of 20 plants) were observed for the bloom and bloomless phenotype. F_3 progeny rows with 100% bloom plants were concluded to have come from F_2 plants homozygous for bm39, whereas F_3 progeny rows segregating for bloom and bloomless plants were concluded to have come from F_2 plants with heterozygous bloom locus. Linkage analysis and map construction were

performed using JoinMap 3.0[®] (Van Ooijen and Voorrips 2001) with a threshold LOD value of 3.0 and a maximum genetic distance of 30 cM. Genetic distances expressed in cM were calculated using the Kosambi mapping function (Kosambi 1944). We further validated the linkage between the bloom locus and the markers using Chi-square test for genetic linkage. The bloom trait data in F_2 , scored as 1 (bloomless) and 2 (bloom) and the marker genotypic data (score as A and B) were used for establishing the genetic linkage using Chi-square test to determine whether two categorical variables are linked and was performed using the Minitab (<http://www.minitab.com>) software (Table 1).

To fine map the bloom locus, we designed an additional 37 SSR primer pairs (Sblm1 to 37 series) between 43 and 50 Mb, for gpsb089 and Xtxp43, on chromosome 1 using sorghum genome sequence v2.1 (https://phytozome.jgi.doe.gov/pz/portal.html#!info?alias=Org_Sbicolor). We tested all of them for polymorphism along with some of the SB, Xtxp, Unnhsbm and msbcIR series SSR markers. To confirm the linkage of bloom locus with these markers, 37 bloomless plants were assayed individually (recessive class analysis: RCA) with polymorphic markers, and Chi-square analysis was carried out to establish the linkage (Table 2). Local maps were prepared for F_2 and F_7 generations of BTx623 $\times$ RS647 with nine and six polymorphic markers, respectively.

Sequencing and data analysis

Altogether four sequencing primers were designed (Table 3) to amplify the whole of candidate gene using the sequence information available at https://phytozome.jgi.doe.gov/pz/portal.html#!info?alias=Org_Sbicolorv2.1.

Table 1 Chi-square analysis for linkage of linked and unlinked polymorphic markers with bm39 locus in the F_2 population of BTx623 $\times$ RS647

Marker	Physical positions (Mb)	Chromosome	Bloomless mean	Bloom mean	Class analysis for bloom and marker segregation				Ch- square value for linkage	P value	
					1-A	1-B	2-A	2-B			
Linked											
Unnhsbm160	14082770	1	1.53	1.86	21	16	24	102	18.76	<0.0001	
gpsb089	43759241	1	1.00	2.00	37	0	0	126	174.61	<0.0001	
Sblm13	44388014	1	1.00	2.00	37	0	0	126	174.61	<0.0001	
Sblm16	44595724	1	1.00	2.00	37	0	0	126	174.61	<0.0001	
SB474	45193765	1	1.00	2.00	37	0	0	126	174.61	<0.0001	
Xtxp43	50203670	1	1.09	2.00	37	0	4	122	148.39	<0.0001	
Xtxp32	52962711	1	1.09	2.00	37	0	4	122	148.39	<0.0001	
Xtxp11	52962828	1	1.18	1.94	30	7	7	199	84.64	<0.0001	
msbcir286	57389044	1	1.09	2.00	37	0	4	122	148.39	<0.0001	
Unlinked											
Xtxp01	61303363	2	1.81	1.74	7	30	35	91	1.23	0.27	
Xtxp285	67805680	3	1.81	1.76	8	29	36	90	0.72	0.39	
Xtxp439	63790155	3	1.8	1.76	8	29	34	92	0.44	0.51	

Table 2 Recessive class analysis in 37 F₂ bloomless plants of BTx623 × RS647

Markers	LG	N ₁ (9.25)	N ₂ (18.5)	N ₃ (9.25)	χ^2	P value
Linked						
Unnhsbm160	1	21	15	1	25.03	3.66E–24
gpsb89	1	37	0	0	109.03	2.11E–24
Sblm13	1	37	0	0	109.03	2.11E–24
Sblm16	1	37	0	0	109.03	2.11E–24
SB474	1	37	0	0	109.03	2.11E–24
Xtxp43	1	37	0	0	109.03	2.11E–24
Xtxp32	1	37	0	0	109.03	2.11E–24
Xtxp11	1	30	7	0	61.83	3.74E–14
msbCIR286	1	37	0	0	109.03	2.11E–24
Unlinked						
Xtxp01	2	9	19	11	0.22	0.89
Xtxp285	3	8	15	13	2.34	0.31
Xtxp439	3	8	17	11	0.6	0.74

N_1 expected number of bloomless individuals with bloomless band (A) if marker is unlinked, N_2 expected number of bloomless individuals heterozygous for the SSR bands from both bloom and bloomless parents (H) if marker is unlinked, N_3 expected number of bloomless individuals with bloom band (B) expected if marker is unlinked

Table 3 Primers used for cloning and sequencing the candidate gene, Sobic.001G269200 (*S. bicolor* v2.1)

S. no.	Primer ID	Physical positions (Mb)	Sequence (5 → 3)	Size (bp)
1	Sblm1008 F	44560718	CCATCTACTGTTATTCCTCCTTGG	550
	Sblm1008 R	44561267	GCACCGTACCTAGAAAGTCCAC	
2	Sblm1009 F	44561031	TTCCACCTCCTCCTCAATATGC	586
	Sblm1009 R	44561616	GCAAACAACATATATAGCACCACC	
3	Sblm1010 F	44561455	CATACCTGAACAAGAACCTCACC	899
	Sblm1006 R	44562353	TCTGGTGTATCTTCTGAGGAAGTC	
4	Sblm1007 F	44562327	GACGACTTCCTCAGAAAGATACACC	573
	Sblm1007 R	44562899	ACTGAAACCCGATTAGAAGAGG	

Both the genotypes BTx623 (bloom) and RS647 (bloomless) were amplified with PCR using four primers. The PCR amplicons were purified using GeneJET purification kit (Thermo Scientific) and cloned into TA cloning vector using puregene Gene-TA cloning kit (Genetix, USA). Positive colonies were identified using blue-white colony screening using Xgal-IPTG in medium. GeneJET Plasmid Mini Prep Kit (Thermo Scientific EU) was used to isolate the plasmids from three positive colonies of each fragment. The whole candidate gene was amplified into four fragments using overlapping primers. The sequencing of cloned fragments was done using universal primer M13 (Sanger sequencing) by Xcelris Labs Limited, Ahmedabad, India. Homology searches were performed by BLASTn algorithm (<http://www.ncbi.nlm.nih.gov/blast>). Overlapping regions from the resulting sequences were removed using BioEdit Sequence Alignment Editor software (<http://www.mbio.ncsu.edu/BioEdit/bioedit.html>). The resultant nucleotide sequences were aligned using the Clustal Omega (<http://www.ebi.ac.uk/Tools/msa/clustalo/>), using coding DNA sequence (CDS) of BTx623 (https://phytozome.jgi.doe.gov/pz/portal.html#!info?alias=Org_Sbicolor) as reference and compared. The nucleotide sequences were converted into peptide sequences by ‘ExPASy’ (<http://web.expasy.org/translate/>). The converted peptide sequences were aligned and compared for any difference or change using ClustalOmega (<http://www.ebi.ac.uk/Tools/msa/clustalo/>).

www.ebi.ac.uk/Tools/msa/clustalo/), using coding DNA sequence (CDS) of BTx623 (https://phytozome.jgi.doe.gov/pz/portal.html#!info?alias=Org_Sbicolor) as reference and compared. The nucleotide sequences were converted into peptide sequences by ‘ExPASy’ (<http://web.expasy.org/translate/>). The converted peptide sequences were aligned and compared for any difference or change using ClustalOmega (<http://www.ebi.ac.uk/Tools/msa/clustalo/>).

Quantitative real-time PCR

Total RNA was extracted from the second top leaf at the boot leaf stage of the crop from two parents BTx623 (B) and RS647 (BL), isogenic RS647C (B) and two near isogenic line sets, NIL 94-1 (B), NIL 94-2 (BL) and NIL 113-1 (B), NIL113-2 (BL) in three replicates. The leaf samples of seven lines were collected in three replicates from tagged plants with a time interval of 5 min. The mean values over replicates were used for comparison. RNA

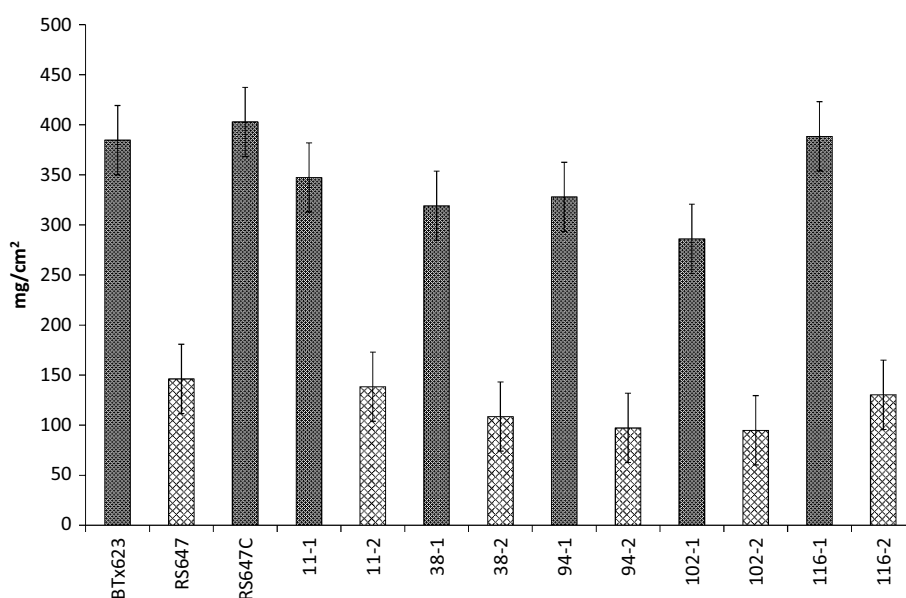
extraction followed the protocol provided by the manufacturer SV Total RNA isolation system kit (Promega corp., USA). First-stranded cDNA synthesis was performed using Thermo Scientific Verso cDNA kit according to the instructions in the manual. To determine the expression pattern of epicuticular wax gene, quantitative real-time (qRT)-PCR was performed using the primer BR3 which was designed using Primer3 (<http://bioinfo.ut.ee/primer3-0.4.0/primer3/>). Ubiquitin was used as reference gene for data normalization (Table 4).

qRT-PCR was carried out in 96-well plates with Light Cycler® 96 system (Roche) using Thermo Scientific Maxima SYBR Green/ROX qPCR master mix kit. Each reaction mix, containing 10 µl of SYBR Green/ROX qPCR master mix, 4 µl diluted cDNA, 1 µl of each forward and reverse primers and 4 µl of water to make a final volume of 20 µl reaction was subjected to the following conditions: 95 °C for 10 min for pre-incubation, followed by 35 cycles of 95 °C for 20 s, 58 °C for 15 s, and 72 °C for 30 s. The melting curve was analyzed at 95 °C for 10 s, 65 °C for 1 min and 97 °C for 1 s after 35 cycles. Each qRT-PCR analysis was performed in triplicate and the mean was used for qRT-PCR analysis (Wang et al. 2013).

Table 4 Primers for RNA expression analysis of Sobic.001G269200

S.No	Primer ID	Sequence (5 → 3)	Size (bp)
1	BR3 F	CTCTAACTCCACTTCCACGTCCAT	138
	BR3 R	GACGATAAGGAGGTATAGGAG GTG	
2	Ubi F	GGTTCGGGAGGTGGCATAGGT	230
	Ubi R	AGCATGTACATTCACGCGGT	

Fig. 2 Mean epicuticular wax load on two parents (BTx623 and RS647), converted line (RS647C) and five sets of NILs measured at the boot leaf stage. Bars indicate standard errors



Results

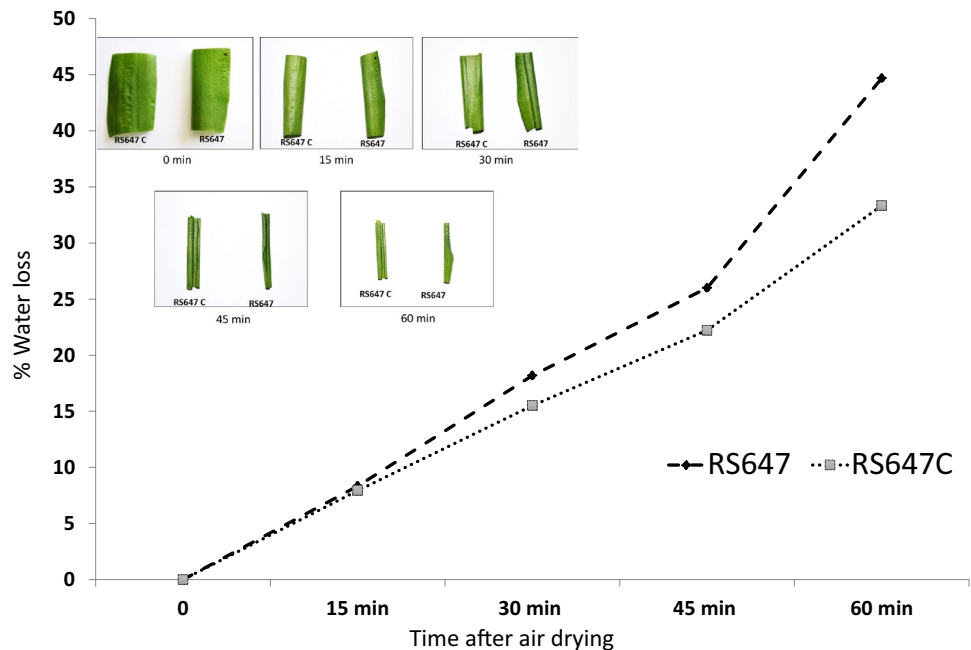
Wax estimation and pattern of water loss

The bloom deposition on the leaf sheath and culm was apparent at the boot leaf stage. The wax loads on leaf were estimated following the colorimetric method in two parents (BTx623 and RS647), converted line (RS647C) and in five sets of NILs (11-1, 11-2; 38-1, 38-2; 94-1, 94-2; 102-1, 102-2; 116-1, 116-2). The mean wax load (of 6 weeks) on bloom parent, BTx623 (385 mg cm⁻²), was significantly higher than the bloomless parent, RS647 (146 mg cm⁻²) (Fig. 2). Similarly, the wax load on the bloom NILs were in general 2.5 times more than their bloomless counterparts. The highest wax load was observed on the converted line RS647C (403 mg cm⁻²) and was almost three times more than its cognate bloomless line, RS647 (146 mg cm⁻²). The pattern of water loss in bloomless RS647 and RS647C over a period of 1 h clearly indicated that the bloomless RS647 had higher water loss (%) per unit time. This loss in water from the leaf surface was 25% higher than RS647C at 60 min (Fig. 3).

Inheritance of epicuticular wax

The bloom deposition on the leaf sheath and culm was apparent at the boot leaf stage. The F₁ plants of all the three crosses, i.e., BTx623 × RS647, 27A × RS647 and 296A × RS647 exhibited bloom deposition suggesting that the bloom allele in bloom parents (BTx623, 27A and 296A) was dominant over the bloomless allele of RS647. The F₂ population of cross P1 had a good fit to the segregation ratio of 3 (126 B):1 (37 BL), indicating dominant

Fig. 3 Mean water loss (%) from the leaf surface in RS647 (bloomless) and its cognate bloom (RS647C) over a period of 1 h



monogenic control of bloom over bloomless ($\chi^2 = 0.46$ and $P = 0.49$). Similarly, 202 plants in P2F₂ segregated as 151(B):51(BL), while 240 plants of P3F₂ segregated as 171(B):69(BL), which gave a good fit to a 3:1 ratio ($\chi^2 = 0.0066$, $P = 0.93$; $\chi^2 = 1.8$ and $P = 0.09$). This also confirmed the monogenic dominant inheritance of the bloom trait. The F_{2,3} progenies of BTx623 × RS647 segregated as 52 homozygous blooms:74 segregating heterozygous:37 homozygous bloomless, as expected for a single gene segregation ratio of 1:2:1 ($\chi^2 = 4.14$ and $P = 0.12$). Similarly, in the F₇ generation, out of 116 F₇ RIL families scored for bloom expression, 61 lines were bloom while 55 were bloomless, again confirming the segregation of 1:1 for a single gene. These results suggest that a difference in a single major gene accounts for the expression of bloom between the parents, and recessive allele causes the bloomless phenotype in RS647.

Targeted mapping of bloom locus

Of the markers linked with epicuticular wax locus (bloom cuticle: *BLMC*) (Burow et al. 2009), six were polymorphic, of which Xsbarslbc10.57 is a genic marker. However, none were found to be significant in BSA. Therefore, we tried other 240 SSR primers distributed over all ten linkage groups of the sorghum genome, of which 25 SSRs were found to be polymorphic between the parents (6 SSRs on SBI-1, 5 on SBI-7, 2 markers on SBI-3 and 5, 3 markers on SBI-2 and 9 and 1 each on SBI-4, 6, 8 and 10). These 25 SSRs were further used in BSA involving bloom and bloomless bulks along with parents. The SSR survey of

bulked extremes identified significant marker from a region on chromosome 1 that was likely to be linked to the bloom locus. An apparent association between the markers and the bloom locus was observed in gpb089 on chromosome 1 with amplification profiles, characteristic of bloom and bloomless parents in the corresponding bulks following bulked segregant analysis.

Out of the new set of SSR markers tested for fine mapping, SSRs SB474, Sblm13 and Sblm16 were polymorphic between the parents and were found to be significant in BSA. Chi-square analysis for linkage of linked and unlinked polymorphic markers with bloom locus in the F₂ population of BTx623 × RS647 clearly revealed the tight genetic linkage on chromosome 1 (Table 1). Further χ^2 test involving 37 bloomless individuals (recessive class analysis) also revealed that the segregation of the markers at the linked loci of bloom deviated significantly from the expected 1:2:1 ratio (for unlinked). χ^2 test for three markers from other chromosome gave a good fit of 1:2:1 segregation ratio, indicating no linkage between the markers and the bloom locus (Table 2).

Map position of bloom locus and its validation

SSR marker data from two mapping populations (163 F₂ plants and 116 F₇ RIL) of BTx623 × RS647 were used for construction of linkage maps using JoinMap 3.0® (Van Ooijen and Voorrips 2001). In the F₂ population, markers Sblm13 and Sblm16 were tightly linked to the bloom locus (bm39) at 0.3 cM on either side (Fig. 4). The F₇ RIL map revealed clustering of the four SSR

markers (SB474, Sblm16, Sblm13, gpsb089) along with the bloom locus at the same point on the linkage map. The analysis clearly established that the bloom locus was located between Sblm13 and Sblm16.

To further confirm the map position of the bloom locus in different genetic backgrounds, co-segregation analysis was carried out in two F_2 populations of crosses, 27A $\times$ RS647 and 296A $\times$ RS647, and local linkage maps were constructed. In both, tight linkage of markers, Sblm13 and Sblm16, with bloom locus was confirmed. In 27A $\times$ RS647 F_2 population, Sblm13 was linked with bloom locus (bm39) at 0.3 cM on one side and Sblm16 was linked at 1.4 cM on the other. Similarly, in the 296A $\times$ RS647 F_2 population, Sblm13 and Sblm16 were linked to the bloom locus (bm39) at 0.3 and 0.1 cM, respectively.

Candidate gene for bloom trait

Linkage analysis delimited bloom locus to a region of 207.89 kb between Sblm13 and Sblm16 markers on chromosome 1 (<https://phytozome.jgi.doe.gov/jbrowse/sorghum>) that is predicted to encode nine sorghum genes (Sobic.001G269100,

Sobic.001G269200, Sobic.001G269300, Sobic.001G269400, Sobic.001G269501, Sobic.001G269600, Sobic.001G269650, Sobic.001G269700 and Sobic.001G269800) (Fig. 5). Of these, Sobic.001G269100 and Sobic.001G269200 are functionally annotated. Sobic.001G269100 codes for ALPHA-MANNOSIDASE and Sobic.001G269200 codes for GDSL-like lipase/acylhydrolase. Sobic.001G269200 was the only gene related to wax synthesis and, therefore, we presumed this as the candidate gene and used it for its RNA expression analysis.

RNA expression analysis

To prove the differential expression of Sobic.001G269200 to indicate its involvement in epicuticular wax synthesis, we studied its expression between two parents (BTx623 and RS647), isogenic RS647C, and in two NIL sets, i.e., 94-1 (B), 94-2 (BL) and 113-1 (B), 113-2 (BL), by qRT-PCR analysis (Fig. 6). Sobic.001G269200 transcript level was significantly higher (fourfold) in bloom plants compared to bloomless plants. The bloom parent BTx623 showed twofold higher transcripts than the bloomless parent RS647. The converted line RS647C showed the highest level of fourfold increase than bloomless RS647.

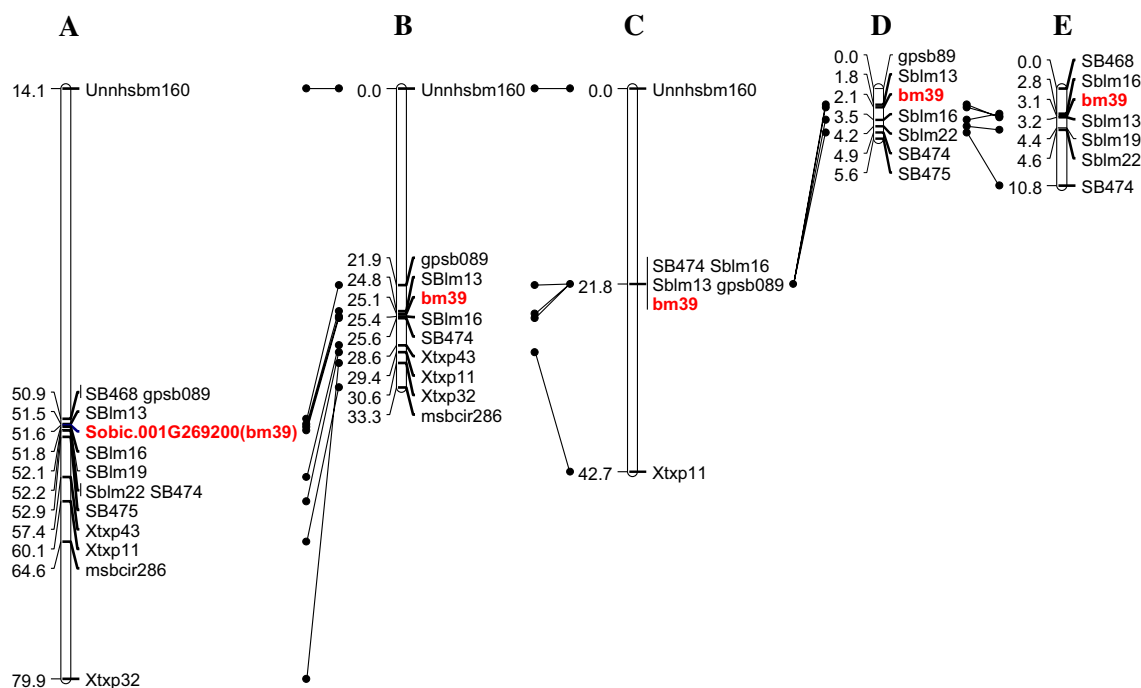
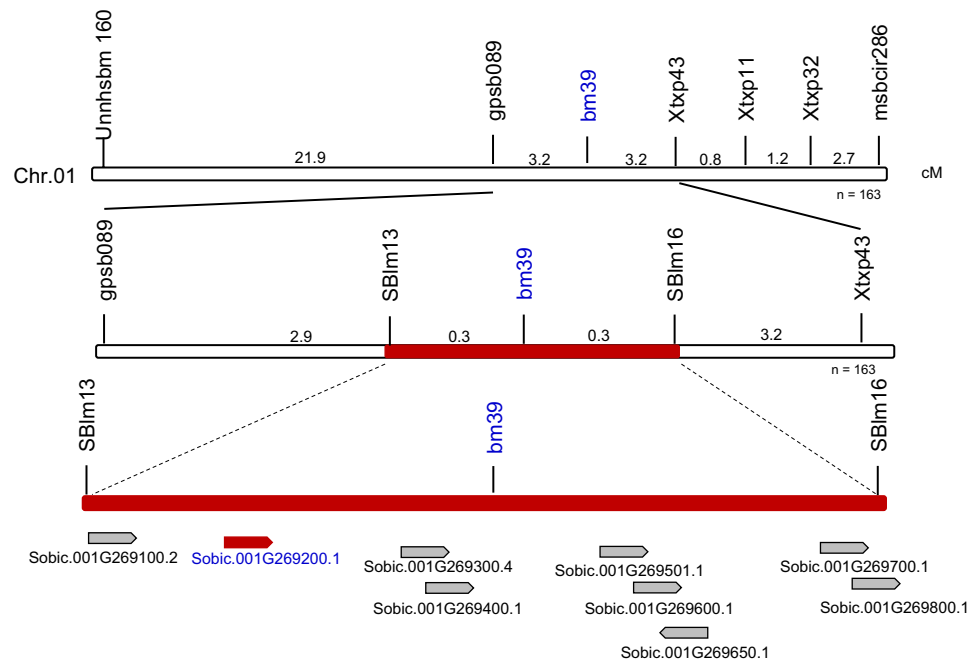


Fig. 4 A Targeted molecular map of sorghum chromosome 1 around the *bm39* locus. The numbers on the left side represent the distances in cM and the corresponding SSR markers are indicated on the right side. The proposed gene for epicuticular wax, *Sobic.001G269200* (*bm39*), is indicated. B Linkage maps of BTx623 $\times$ RS647 F_2 map-

ping population, C BTx623 $\times$ RS647 F_7 RIL population. Validation of linked primers and their corresponding linkage maps in D 27A $\times$ RS647 F_2 mapping population, E 296A $\times$ RS647 F_2 mapping population

Fig. 5 A fine-scale map of the *bm39* locus on sorghum chromosome 1 with candidate genes. Most probable candidate gene *Sobic.001G269200* (*bm39*) is indicated



Sequence analysis of coding region and peptides

We designed four pairs of sequencing primers to amplify the full length of candidate gene (*S. bicolor* v2.1) in two parents, BTx623 and RS647. In both, cloned sequence was of 2182 bp length. The gene has four exons and three introns and translated protein contained 316 aa. Sequences of the cloned BTx623 and the reference BTx623 (https://phytozome.jgi.doe.gov/pz/portal.html#!info?alias=Org_Sbicolor) were the same. However, parents, RS647 and BTx623, differed for two ‘G’ to ‘C’ transversions, which were observed in bloomless RS647. One transversion was positioned at the 297th bp, while the second ‘G’ to ‘C’ transversion was found at the 971th bp (Fig. 7). When

parents were compared for their exon regions of 948 bp length, we could observe only the second ‘G’ to ‘C’ at 371 bp, falling within the genic sequence.

Predicted peptide sequences of both parents consist of 316 aa, like the reference sequence of BTx623 (https://phytozome.jgi.doe.gov/pz/portal.html#!info?alias=Org_Sbicolor). We did not find any difference between the peptide sequence of reference BTx623 and predicted peptide sequence of BTx623. However, between two parents (BTx623 and RS647) at the 124th aa position, non-synonymous mutation was observed wherein threonine (T) replaced serine (S) in RS647 sequence (Fig. 8). This change in amino acid corresponded well with the ‘G’ to ‘C’ transversion observed at the 371th position in the exon sequence. In BTx623, only one alpha helix spanning at position 190–195 aa of predicted peptide sequence was available. However, in RS647, in addition to alpha helix at position 190–195 aa, one more alpha helix was noticed at position 185–188 aa in the predicted sequence.

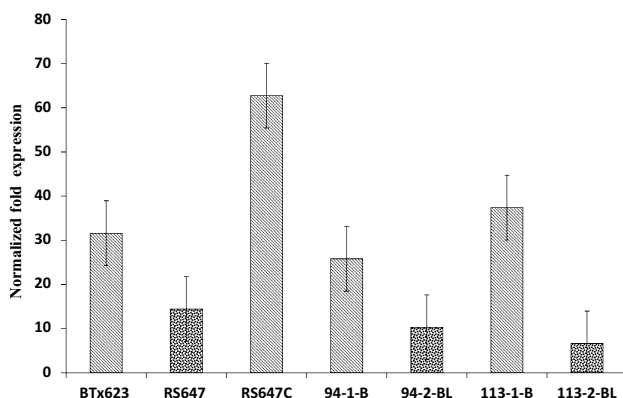


Fig. 6 qRT-PCR analysis of *Sobic.001G269200* (*bm39*) gene in bloom (B) and bloomless (BL) genotypes. Bars indicate standard errors

Discussion

Plant epicuticular wax layer provides the first line of defense between the plant and its environment, involving biotic and abiotic stress factors. Among the cereal crops, sorghum probably produces the highest amounts of epicuticular wax (Burow et al. 2008), visually recognized as the waxy bloom on its abaxial leaf surface and on culm sheath (Blum 2004). Wax depositions take an amorphous, a flaky, or a “starlike” shape on the leaf lamina, while they form

Fig. 7 Sequence comparison between BTx623 (bloom) and RS647 (bloomless) genotypes. Start of the gene is indicated with an *arrow*. Sequence with 2 bp changes are indicated with *inverted arrow*. Only a part of the gene with observed changes from G (in BTx623) to C (in RS647) is depicted

BTx623	CCTTCTAGTCTCCACCACGGTTAGGGGCACCACTTCGGTCTCCTCTTTTTCCTCCGCGG	300
RS647	CCTTCTAGTCTCCACCACGGTTAGGGGCACCACTTCGGTCTCCTCTTTTTCCTCCGCGG	300

BTx623	CACACTCCTGTTCTTCCACCTCCTCCTCAATATGCGGCCGATCCTCCTCCTCCTCCACCA	360
RS647	CACACTCCTGTTCTTCCACCTCCTCCTCAATATGCGGCCGATCCTCCTCCTCCTCCACCA	360

BTx623	CCTATGGGTGGGCCCAAGTGCCAGCTCTGATTGCGTTTGGAGACTCAATCGTGAGACACC	420
RS647	CCTATGGGTGGGCCCAAGTGCCAGCTCTGATTGCGTTTGGAGACTCAATCGTGAGACACC	420

BTx623	GGCAACAATAACTACCTCATGACTGTTGTGAAGGCCAACTTCCCGCGTACGGCAAGGAG	480
RS647	GGCAACAATAACTACCTCATGACTGTTGTGAAGGCCAACTTCCCGCGTACGGCAAGGAG	480

BTx623	TACCTTGGCCACAAGGCCACCGGCAGGTTCTCCGACGCAAGATCACCGTGGACTTTCTA	540
RS647	TACCTTGGCCACAAGGCCACCGGCAGGTTCTCCGACGCAAGATCACCGTGGACTTTCTA	540

BTx623	GGTACGGTGC GGATGTGTTTAACTTTGTTTTTACATCAAAATCTTGAATTTCTGATGA	600
RS647	GGTACGGTGC GGATGTGTTTAACTTTGTTTTTACATCAAAATCTTGAATTTCTGATGA	600

BTx623	GGCATCAGGCATATGGTAATTAACACCACCTGACATTACGAATGGCTCTTTATATATAC	660
RS647	GGCATCAGGCATATGGTAATTAACACCACCTGACATTACGAATGGCTCTTTATATATAC	660

BTx623	GTATATAATTAACGGGCCGGTAAATACCTTATCTCTATCATCAGCGTCTGCGTTTGGGTT	720
RS647	GTATATAATTAACGGGCCGGTAAATACCTTATCTCTATCATCAGCGTCTGCGTTTGGGTT	720

BTx623	GAAGGAGACGCTCCCGCCATACCTGAACAAGAACCTCACCTGGAGGACCTCAAGACCGG	780
RS647	GAAGGAGACGCTCCCGCCATACCTGAACAAGAACCTCACCTGGAGGACCTCAAGACCGG	780

BTx623	CGTCAGCTTCGCGTCCGCGGGGAGCGGCTACAACAACGCCAGTGCCGGACGTCGGTGAG	840
RS647	CGTCAGCTTCGCGTCCGCGGGGAGCGGCTACAACAACGCCAGTGCCGGACGTCGGTGAG	840

BTx623	CTAGCAGACCTCGATAGAAATATTACTATAATAGGTGGTGGTGCTATATATGTTGTTGCA	900
RS647	CTAGCAGACCTCGATAGAAATATTACTATAATAGGTGGTGGTGCTATATATGTTGTTGCA	900

BTx623	TGCGTGCAGTCGACCATGACGATCGAGCGGCAGCTGCAGCTGTTCTCCGAGTACAAGGCG	960
RS647	TGCGTGCAGTCGACCATGACGATCGAGCGGCAGCTGCAGCTGTTCTCCGAGTACAAGGCG	960

BTx623	AAGGTGGGGAACATCCCCGAGAGGGCCCTCTTCGTGGTCTGCTCCGGCAGCAACGACATC	1020
RS647	AAGGTGGGGAACATCCCCGAGAGGGCCCTCTTCGTGGTCTGCTCCGGCAGCAACGACATC	1020

a dense, thin fiber-like mass on the sorghum leaf sheath (Blum 1975; Tarumoto et al. 1981). The bloom layer serves as a potentially important drought avoidance mechanism in sorghum (Jordan et al. 1983, 1984).

The bloom parent BTx623 had approximately four-and-half times more wax (384.6 mg cm^{-2}) than the bloomless line RS647 (146.2 mg cm^{-2}). The converted isogenic bloom RS647C had threefold more wax deposition compared to its bloomless RS647. As expected, the water loss in bloomless RS647 was higher than in its cognate RS647C, clearly implying the significance of epicuticular wax in conserving water in the plant. Significant epicuticular wax reduction or possible elimination could result in significant increase in nighttime transpiration, which could be equivalent to 30% of daytime water loss (Burow et al. 2008). The epicuticular wax load is probably one

important factor that could contribute to the greater water efficiency and drought tolerance of sorghum. Wax-deficient mutants in Arabidopsis, rice and Brassica generally display increased chlorophyll leaching, rate of water loss, and sensitivity to drought (Chen et al. 2003; Islam et al. 2009; Liu et al. 2014). Thus, it is very apparent that the role of epicuticular wax load in water use efficiency and drought tolerance needs to be studied, especially in the context of climate change and shortage of water supply. Higher wax deposition in bloom over bloomless mutants was reported in sorghum (Burow et al. 2008, 2009; Ebercon et al. 1977; Jordan et al. 1984). Lee et al. (2015) reported 1.9-fold more wax content in the bloomed broccoli than in the bloomless, while Zhang et al. (2013) reported nearly threefold higher wax in bloom than that in the bloomless genotype in Chinese cabbage.

BTx623	MGGPKVPALIAFGDSIVDTGNNNYLMTVVKANFPYPYKGKEYPGHKATGRFSDGKITVDFLA	60
RS647	MGGPKVPALIAFGDSIVDTGNNNYLMTVVKANFPYPYKGKEYPGHKATGRFSDGKITVDFLA	60

BTx623	SAFGLKETLPPYLNKNLTLEDLKTGVSFASAGSGYNNATCRTSSTMTIERQLQLFSEYKA	120
RS647	SAFGLKETLPPYLNKNLTLEDLKTGVSFASAGSGYNNATCRTSSTMTIERQLQLFSEYKA	120

↓		
BTx623	KVG S IPERALFVVCSGSNDIVEHFTLADSMTSPEYAEMMARRAIGLVEALIGQGARQIAL	180
RS647	KVG T IPERALFVVCSGSNDIVEHFTLADSMTSPEYAEMMARRAIGLVEALIGQGARQIAL	180
:**		
BTx623	TGAPPVGCVPSPQRRIAGGVRTQCATDRNQLALLFNKVSLEVAKLSGKYRGVNIFYVDLY	240
RS647	TGAPPVGCVPSPQRRIAGGVRTQCATDRNQLALLFNKVSLEVAKLSGKYRGVNIFYVDLY	240

BTx623	SIVADVQRYQDLGFKDGKDACCYIGLAVGPLCNVGSRTCPDPSKYVFWDSYHPTERAY	300
RS647	SIVADVQRYQDLGFKDGKDACCYIGLAVGPLCNVGSRTCPDPSKYVFWDSYHPTERAY	300

BTx623	KIMIDDFLRRYTRYIH	316
RS647	KIMIDDFLRRYTRYIH	316

Fig. 8 Predicted amino acid sequence comparison between BTx623 (bloom) and RS647 (bloomless). Change in RS647 is indicated with an inverted arrow

Bloom layer deposition is genetically controlled, wherein bloomless phenotype significantly reduces the bloom load on the plant. All three F_1 plants of crosses BTx623 $\times$ RS647, 296A $\times$ RS647, 27A $\times$ RS647 exhibited heavy bloom deposition, indicating the dominance of bloom over the bloomless phenotype. The segregation pattern of bloom and bloomless plants in each of the F_2 population was observed to be the same. F_2 plants of three crosses segregated with a monogenic ratio of 3 (bloom):1 (bloomless), which suggested the existence of single nuclear gene difference between the bloom parents (BTx623, 296A and 27A) and the recessive bloomless parent, RS647. Thus, a mutant bloomless condition was determined by the recessive allele of a single gene in RS647. The recessivity of bloomless phenotype to bloom was reported earlier in sorghum (Ayyangar and Ponnaiya 1941; Burow et al. 2008; Peterson et al. 1979).

Genetic mapping using molecular markers will facilitate the identification of the position of a genetic locus on the chromosome. In sorghum, though several bloomless mutants and their allelic relations have been reported (Burow et al. 2008; Jenks et al. 1994; Peters et al. 2009; Punhuri et al. 2017), molecular map positions of only two bloomless mutant loci, *BLMC* (bloom cuticle), and 2I-400-2-11-3 on chromosome 10 and 6, respectively, are reported (Burow et al. 2009; Mizuno et al. 2013). In a recent study, bm2 and *BLMC* that affect epicuticular wax deposition in sorghum are found to be allelic (Punhuri et al. 2017) which was also reported to be allelic to bm6 (bm2-2), bm22

(bm2-3), bm33 (bm2-4), Txbm1 (bm2-5), CK60bm (bm2-6) (Peters et al. 2009). Hence, bm2 locus is on chromosome 10.

Though epicuticular wax is a quantitative trait with many genes involved in its biosynthesis, in this study we undertook a strategy to map a major locus that caused the absence of epicuticular wax (bloomless) in a natural bloomless mutant line, RS647, available in our sorghum program. Our strategy on phenotyping the segregating population on a binary scale (presence/absence of bloom), though, resulted in identifying a major locus for epicuticular wax synthesis, but we do agree that there could be many other genes involved in the expression of bloom trait with minor effects, which our strategy would not identify. Additional studies on phenotypic quantification of wax load and its analysis through QTL approach might detect other regions on sorghum genome involved in epicuticular wax biosynthesis. Our strategy for locating major locus for bloomless mutant through linkage analysis clearly established that the present bloom locus was located between SSRs, Sblm13 and Sblm16 on chromosome 1, which were tightly linked at 0.3 cM on either side. Our validation studies involving different bloom parents (27A and 296A) also confirmed the map position on chromosome 1. Consistent with the earlier nomenclature for bloom loci (Peters et al. 2009; Peterson et al. 1982), we designated the new bloomless locus as bm39. We could not conduct allelism test between bm39 and various bloomless mutants reported (Burow et al. 2009; Jiao et al. 2016; Mizuno et al. 2013;

Peters et al. 2009), as representative genetic stocks were not available with us. It is possible that bm39 may be allelic to one of the bm mutants reported earlier.

Considering map positions of bm39 on chromosome 1, *BLMC* on chromosome 10 (Burow et al. 2009) and 2I-400-2-11-3 on chromosome 6 (Mizuno et al. 2013), we strongly believe that these bloomless mutants are due to mutation in different genes of epicuticular wax synthesis. Therefore, the current bm39 locus on chromosome 1 for bloomless phenotype is new and is different from the *BLMC* (bm2) and 2I-400-2-11-3. We expect that the complementation test between these bloomless mutants should restore the wild bloom phenotype, which needs to be confirmed.

Our study delimits the new bloom locus to a genomic region of 207.89 kb on chromosome 1. There are nine predicted genes in the target region. Based on the sequence annotation (<https://phytozome.jgi.doe.gov/jbrowse>), the most likely candidate was Sobic.001G269200, GDSL-like lipase/acylhydrolase, as it could have direct role in the metabolism of the fatty acid composition of the sorghum wax. GDSL lipases have received significant interest over the past few years. They are hydrolytic enzymes that exhibit a broad range of substrate specificity and carry out several enzymatic reactions, including thioesterase, protease, lysophospholipase and carboxypeptidase activities (Akoh et al. 2004). These proteins with their specific expression in epidermal cells of several species at early stages of development indicated their possible involvement in cutin synthesis (Mintz-Oron et al. 2008; Reina et al. 2007; Shi et al. 2011; Yeats et al. 2010). In recent studies, strong evidence has accumulated showing that GDSL protein is essential for cutin synthesis and deposition. Two studies, using different experimental approaches, reported independently the role in cutin synthesis of a GDSL lipase and concluded that GDSL lipase is extracellular and locates to the cutinized region of the outer epidermal cell wall of developing tomato fruits (Girard et al. 2012; Yeats et al. 2012). Two GDSL-motif lipases were shown to be associated with cuticle assembly, namely BODYGUARD in Arabidopsis (Kurdyukov et al. 2006) and AgaSGNH in *Agave americana* L. (Reina et al. 2007). Both proteins were shown to be putatively involved in the polymerization process that occurs on the extracellular surface of epidermal cells.

Gene Sobic.001G269200 (Chr01:44560907.0.44562775) encoding GDSL-like lipase/acylhydrolase of the present study is most likely to be involved in the epicuticular wax deposition on the leaf and culm surface in sorghum. Sobic.010G001900 (Sb10g000390.1) of Burow et al. (2009), annotated for acyl-CoA oxidase involved in lipid metabolism, was the potential candidate gene for bloomless phenotype. The present potential candidate gene, Sobic.001G269200, shows protein homology to the maize gene GRMZM2G390678 (Chr05:40495144-40497737),

rice gene LOC_Os10g05088.1 (Chr10:2492588.0.2496864) and wheat gene Traes_6AS_B3D6765FB (ta_iwgs_6as_v1_4368834:5908.0.7265) with the same functional annotation (<https://phytozome.jgi.doe.gov/pz/portal.html#>). We also looked for the presence of related genes in sorghum genome based on protein homology. For our candidate Sobic.001G269200, 164 sorghum genes were identified with a % identity ranging from 23 to 81. Only two genes, Sobic.002G414300 and Sobic.001G269800, showed 81% identity. Similarly, for Sobic.010G001900 of Burow et al. (2009), 14 genes were found with % identity varying from 24 to 93, of which Sobic.010G049100 had maximum % identity.

Wax-deficient mutants and their genes affecting cuticular wax biosynthesis, its transport, transcriptional regulation and post translation regulation have been reported in many species (Supplementary Table 3). Mutants with deficient or altered wax coatings, identified by their non-glaucous or glossy appearance, have been reported in Arabidopsis, maize, barley and rape (Kunst and Samuels 2003). Studies of the eceriferum (*cer*) mutants in Arabidopsis and glossy (*gl*) mutants in maize and rice led to the identification and isolation of a number of wax-related genes (Gan et al. 2016; Li et al. 2013; Samuels et al. 2008). In Arabidopsis, loss of function of genes for the four ER-bound enzymes involved in fatty acid elongation, such as KCS6/CTU1/CER6 (Millar et al. 1999), KCS2/DAISY (Lee et al. 2009), KCS9 (Kim et al. 2013), KCS1 (Todd et al. 1999), KCR1 (Beaudoin et al. 2009), HCD/PAS2 (Bach et al. 2008) and ECR/CER10 (Zheng et al. 2005), consistently resulted in decreased levels of cuticular wax. Several genes in rice, WSL1, a homolog of Arabidopsis KCS (Yu et al. 2008), and three homologs of Glossy1 rice genes, viz. WSL2/GL1-1, GL1-3 and GL1-6 (Mao et al. 2012; Qin et al. 2005; Zhou et al. 2013, 2015), are involved in epicuticular wax biosynthesis. In wheat, the wax on leaves and stems was mainly controlled by two sets of genes: glaucousness loci (*W1* and *W2*) and non-glaucousness loci (*Iw1* and *Iw2*) (Lu et al. 2015).

The GDSL esterase and lipase families play important roles in abiotic stress, pathogen defense, seed development and lipid metabolism (Tan et al. 2014). The presence of the bloom layer in sorghum is associated with resistance to green bug (Weibel and Starks 1986) and fungal pathogen, *E. turcicum* (Jenks et al. 1994), sheath blight (Kasuga et al. 2001). Oh et al. (2005) suggested that GDSL-like motif (GLIP1) protein, via ethylene signaling, could be a critical component in plant resistance to the necrotrophic fungus *Alternaria brassicicola* in Arabidopsis.

Our qRT-PCR analysis indeed confirmed the significantly reduced levels of Sobic.001G269200 transcript in bloomless genotypes against the bloom. Further experiments on co-segregation analysis may prove the

involvement of Sobic.001269200 in the expression of bloom/bloomless phenotype in sorghum. The sequence comparison in the epicuticular wax gene from bloom and bloomless genotypes suggests a very high degree (99.85%) of conserved sequence, i.e., 1365 bp out of 1367 bp conserved. Both serine and threonine amino acids contain the same OH group in the side chains, which indicates that there is no great variation in reactivity of side chain in terms of protein folding or in its function. However, the additional CH₂ group present in the side chain of threonine may affect the protein function. Simultaneously, the folding of both these peptides was predicted using relevant bioinformatics software's, due to unavailability of a known 3D structure of this protein and the absence of $\geq 80\%$ homology containing protein/peptide sequence in available 3D structure source. Apart from the above, the low quantity of mRNA identified in expression analysis might have translated less number of this protein molecules and ultimately led to less accumulation of wax in RS647, which is a bloomless genotype. Between the two transversions noticed, considering the expression and phenotypic analysis, transversion in the non-translating region (5'UTR) (371 bp) appears to be responsible for bloomless situation over the transversion in the coding region that requires further biochemical studies.

The incidence of two nucleotide differences between the waxy and non-waxy alleles within the coding region was reported in sorghum (McIntyre et al. 2008). The authors concluded that the glutamine to histidine substitution noticed at the position corresponding to one of the SNPs was likely responsible for producing the inactive form of GBSS resulting in the non-waxy phenotype. Another such case of an SNP causing alanine to threonine substitution that resulted in the production of an inactive GBSS was also reported in wheat (Yanagisawa et al. 2001). While these two cases of SNP changes were reported to be within the coding region, changes in the adjacent non-coding regions as being responsible for the non-production of the GBSS or production of an inactive GBSS was indicated in sorghum (Hamblin et al. 2007). Konishi et al. (2006) suggested an SNP 12 kb away from the ORF to be responsible for loss of shattering in japonica cultivars of rice.

Application in sorghum breeding

Like in other crops, epicuticular wax in sorghum plays different roles. Sorghum is known for its adaptation to moisture stress and acts as a model crop for elucidation of mechanisms of drought tolerance. The presence of epicuticular wax reduces non-stomatal water loss and enhances water use efficiency of sorghum by regulating nighttime water loss (Burow et al. 2009). Sorghum is also known

as an important high biomass cereal crop preferred for its nutritious green and dry forage. Reduced epicuticular wax load, as achieved by the bloomless genotype, significantly improved the estimated forage digestibility by ruminants (Cummins and Dobson 1972; Cummins and Sudweeks 1976). Perhaps, the isogenic bloom and bloomless RS647 and NIL sets developed in the program may provide a system for further examination of the role of epicuticular wax in drought tolerance, forage digestibility and pest resistance. Mutant brown midrib phenotype in sorghum was associated with altered lignin content/composition and increased in vitro digestibility (Akin et al. 1986; Porter et al. 1978). Probably, sorghum with bloomless and brown midrib double mutant may significantly contribute to improve forage quality in sorghum, which needs to be studied further.

Author contribution statement RM: designed the experiments; AU and PM: conducted the experiments; RM: analyzed the data and wrote the manuscript; YV, VAT: discussed the analysis and assisted in manuscript preparation. All authors approved the final version of the manuscript.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no competing interests.

Ethical standards The authors declare that no ethical standards have been violated in the course of the study.

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